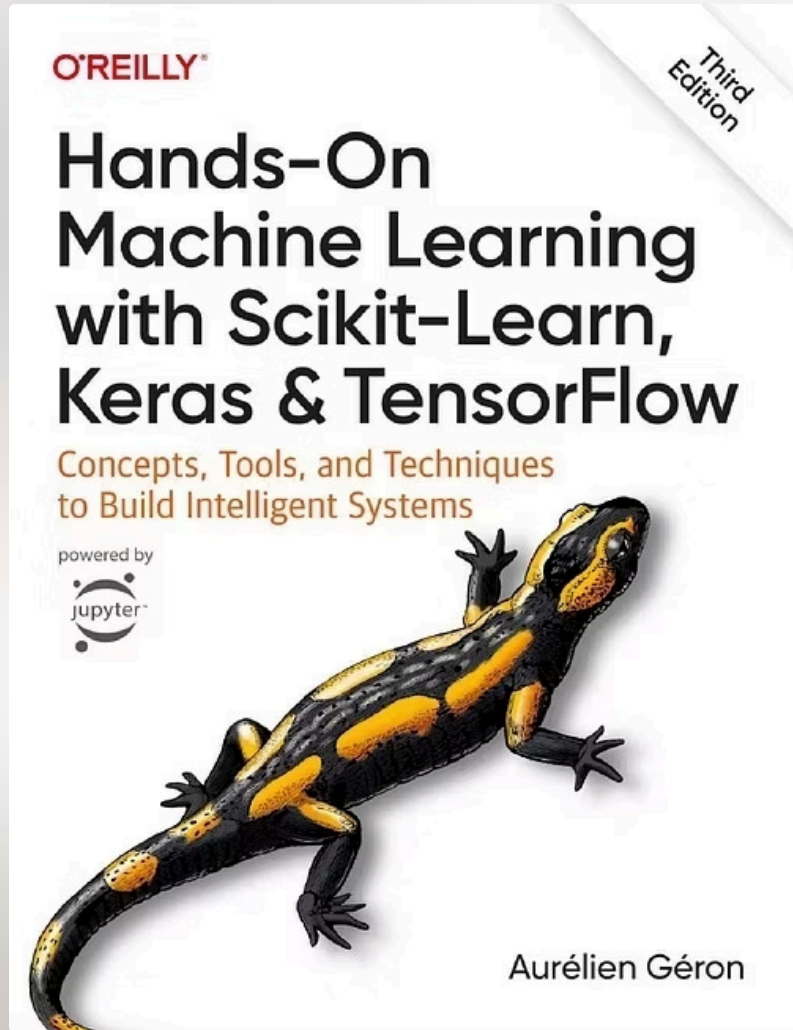




Machine Learning

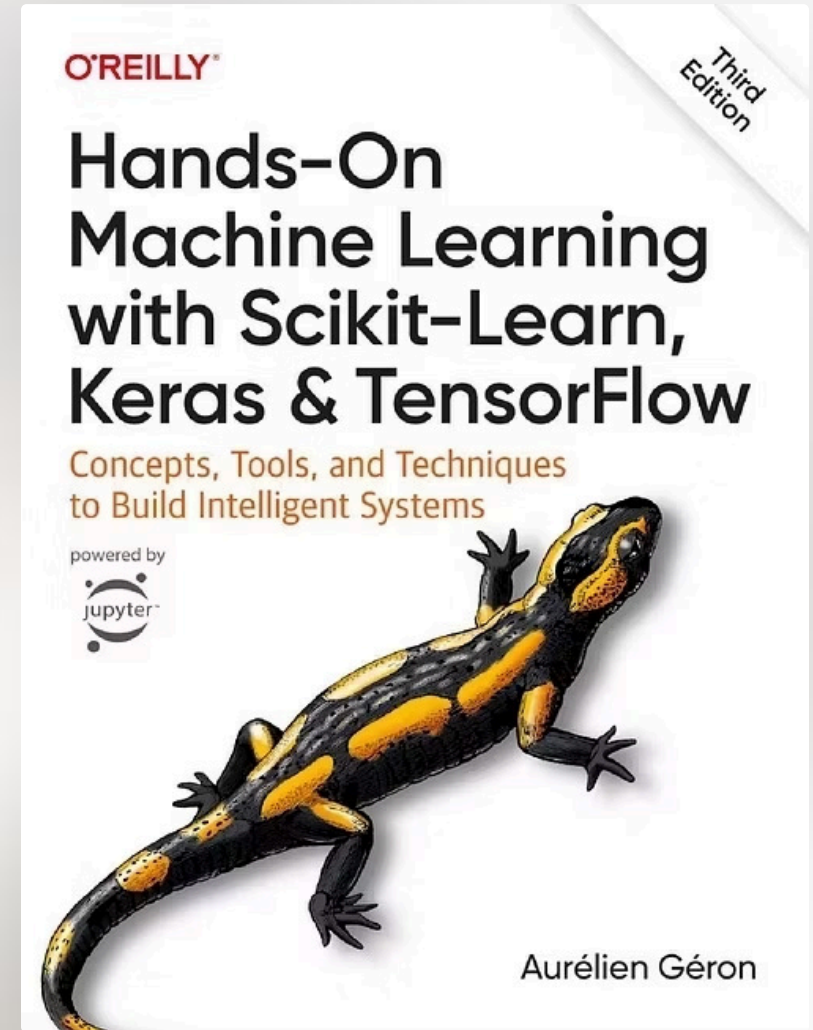
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 07/08/2025

Machine Learning

Chapter 1 – The Machine Learning Landscape





Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
- 4. Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
 - **Classification criteria #1: Supervision during training**
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and **Reinforcement Learning**
 - **Classification criteria #2: Approach to Learning Method**
 - Batch and Online (Incremental) Learning
 - **Classification criteria #3: Learning Method**
 - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - Underfitting the Training Data
 - Stepping Back
6. **Testing and Validating**
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch



Classification criteria #1: Supervision during training

Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning

Reinforcement Learning: Overview

- The learning system is called an agent.
- The agent observes its environment, selects and performs actions, and receives rewards or penalties.

Goal:

- Learn the best strategy, called a policy, to maximize rewards over time.



Reinforcement Learning

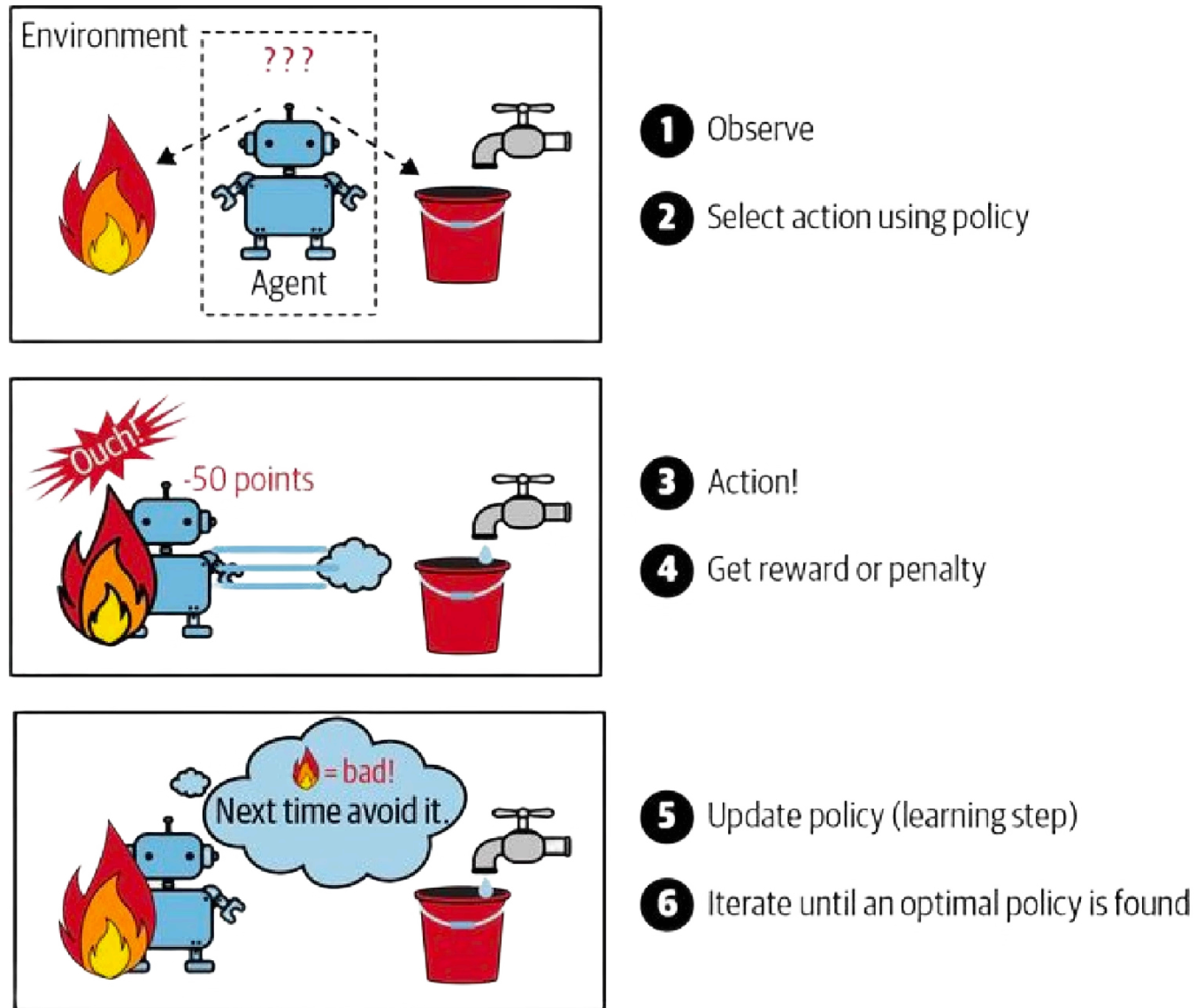


Figure 1-13. Reinforcement learning

How Reinforcement Learning Works

Agent's Actions:

- The agent performs actions based on the **current state** of the **environment**.
- Each **action results** in a **reward or penalty**.

Learning Process:

- The agent **learns which actions** lead to the **most reward** over time.
- The **strategy to select actions** is known as the **policy**.





Example: Robots Learning to Walk



Robots:

Reinforcement learning algorithms **help robots learn complex tasks** like walking.



Goal:

The robot **adjusts its actions to achieve the best performance** (e.g., maintaining balance or walking efficiently).

Example: DeepMind's AlphaGo



AlphaGo's Achievement:

- Beat Ke Jie, the number one ranked player in the world, in May 2017.
- The agent learned its winning policy by analyzing millions of games and playing against itself.



Offline Learning:

- Learning was turned off during the games against the champion;
 - **Offline Learning**
- AlphaGo was applying the learned policy.



Key Takeaways on Reinforcement Learning



Agent and Environment:

The **agent** learns from interaction with the **environment** through **rewards** and **penalties**.



Policy:

The **agent** learns what **actions** to take in **different situations** to maximize long-term **rewards**.

Next: Batch Learning



Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts